

Basics of Statistics

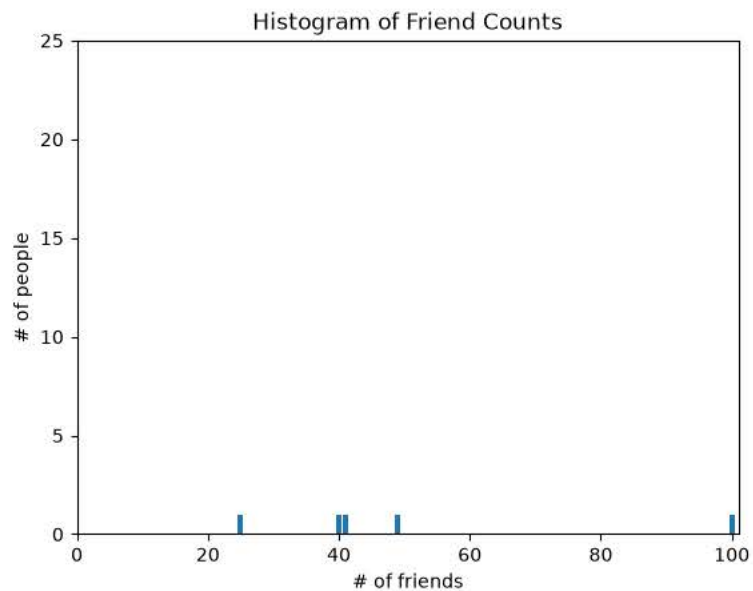
Definition: Science of collection, presentation, analysis, and reasonable interpretation of data.

Statistics presents a rigorous scientific method for gaining insight into data. For example, suppose we measure the weight of 100 patients in a study. With so many measurements, simply looking at the data fails to provide an informative account. However statistics can give an instant overall picture of data based on graphical presentation or numerical summarization irrespective to the number of data points. Besides data summarization, another important task of statistics is to make inference and predict relations of variables.

Statistical Description of Data

- ❑ Statistics describes a numeric set of data by its
 - ❑ Center
 - ❑ Variability
 - ❑ Shape
- ❑ Statistics describes a categorical set of data by
 - ❑ Frequency, percentage or proportion of each category

```
[1]: from matplotlib import pyplot as plt
    from collections import Counter
    num_friends = [100, 49, 41, 40, 25]
    friend_counts = Counter(num_friends)
    xs = range(101)
    ys = [friend_counts[x] for x in xs]
    plt.bar(xs, ys)
    plt.axis([0, 101, 0, 25])
    plt.title("Histogram of Friend Counts")
    plt.xlabel("# of friends")
    plt.ylabel("# of people")
    plt.show()
```



o Mean : (average)

- o The "Mean" is computed by **adding** all of the numbers in the data together and **dividing** by the number elements contained in the data set.

◌ Example :

◌ Data Set = 2, 5, 9, 3, 5, 4, 7

◌ Number of Elements in Data Set = 7

◌ Mean = $(2 + 5 + 9 + 7 + 5 + 4 + 3) / 7 = 5$

o **Median : (middle)**

- o The "Median" of a data set is dependent on whether the number of elements in the data set is odd or even.
- o First reorder the data set from the smallest to the largest
- o Mark off high and low values until you reach the **middle**.
- o If there 2 middles, add them and **divide** by 2.

Examples : Odd Number of Elements

o Data Set = 2, 5, 9, 3, 5, 4, 7

Reordered = 2, 3, 4, 5, 5, 7, 9

^

Median = 5

Examples : Even Number of Elements

Data Set = 2, 5, 9, 3, 5, 4

Reordered = 2, 3, 4, 5, 5, 9
 ^ ^

$$\text{Median} = (4 + 5) / 2 = 4.5$$

Mode : (most often)

- The "Mode" for a data set is the element that occurs the most often.
- It is not uncommon for a data set to have more than one mode.
- This happens when two or more elements occur with equal frequency in the data set.

o **Example :**

o **Data Set** = 2, 5, 9, 3, 5, 4, 7

o **Mode** = 5

o **Example:**

o **Data Set** = 2, 5, 2, 3, 5, 4, 7

o **Modes** = 2 and 5